6. The demand and supply functions are given by $D(x)=16-x^2$ *and* $S(x)=2x^2+4$ are under perfect competition, then the equilibrium price x is

(a) 2 (b) 3 (c) 4 (d) 5

7. The marginal revenue and marginal cost functions of a company are $MR=30-6x$ and $MC=-24+3x$ where x is the product, then the profit function is

(a) $9x^2+54x$ (b) $9x^2-54x$

(c) $54x-\frac{9x^2}{2}$ (d) $54x-\frac{9x^2}{2}+k$

8. The given demand and supply function are given by $D(x)=20-5x$ *and* $S(x)=4x+8$ if they are under perfect competition then the equilibrium demand is

(a) 40 (b) $\frac{41}{2}$ (c) $\frac{40}{3}$ (d) $\frac{41}{5}$

9. If the marginal revenue $MR=35+7x-3x^2$, then the average revenue AR is

(a) $35x+\frac{7x^2}{2}-x^3$ (b) $35+\frac{7x}{2}-x^2$

(c) $35+\frac{7x}{2}+x^2$ (d) $35+7x+x^2$

10. The profit of a function $p(x)$ is maximum when

(a) $MC-MR=0$ (b) MC=0

(c) MR=0 (d) MC+MR=0

11. For the demand function $p(x)$, the elasticity of demand with respect to price is unity then

(a) revenue is constant

(b) cost function is constant

(c) profit is constant

(d) none of these

12. The demand function for the marginal function $MR=100-9x^2$ is

(a) $100-3x^2$ (b) $100x-3x^2$

(c) $100x-9x^2$ (d) $100+9x^2$

13. When $x_0=5$ *and* $p_0=3$ the consumer's surplus for the demand function $p_d=28-x^2$ is

(a) 250 units (b) $\frac{250}{3}$ *units*

(c) $\frac{251}{2}$ *units* (d) $\frac{251}{3}$ *units*

14. When $x_0=2$ *and* $P_0=12$ the producer's surplus for the supply function $P_s=2x^2+4$ is

(a) $\frac{31}{5}$ *units* (b) $\frac{31}{2}$ *units*

(c) $\frac{32}{3}$ *units* (d) $\frac{30}{7}$ *units*

15. Area bounded by $y=x$ between the lines $y=1, y=2$ with $y=axis$ is

(a) $\frac{1}{2}$ sq.units (b) $\frac{5}{2}$ sq.units

(c) $\frac{3}{2}$ sq.units (d) 1 sq.unit

16. The producer's surplus when the supply function for a commodity is $P=3+x$ *and* $x_0=3$ is

(a) $\frac{5}{2}$ (b) $\frac{9}{2}$ (c) $\frac{3}{2}$ (d) $\frac{7}{2}$

17. The marginal cost function is $MC=100\sqrt{x}$. find AC given that TC =0 when the out put is zero is

(a) $\frac{200}{3}x^{\frac{1}{2}}$ (b) $\frac{200}{3}x^{\frac{3}{2}}$

(c) $\frac{200}{3x^{\frac{3}{2}}}$ (d) $\frac{200}{3x^{\frac{1}{2}}}$

18. The demand and supply function of a commodity are $P(x)=(x-5)^2$ and $S(x)=x^2+x+3$ then the equilibrium quantity x_0 is

(a) 5 (b) 2 (c) 3 (d) 1

19. The demand and supply function of a commodity are $D(x)=25-2x$ and $S(x)=\frac{10+x}{4}$ then the equilibrium price P_0 is

(a) 5 (b) 2 (c) 3 (d) 10

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